Unsmoothed Real Estate Returns

To illustrate the effects of unsmoothing, Figure 13.3 plots direct real estate returns from the National Council of Real Estate Investment Fiduciaries (NCREIF), which constructs an institutional property index from data reported by its members.[18] Because this is an appraisal index, NCREIF real estate returns are highly autocorrelated. From March, 1978 to December, 2011, the first-order autocorrelation of NCREIF returns is 0.78. The raw reported data is shown in the solid line. I graph unsmoothed returns in the squares applying the filter of equations (13.1) and (13.2). All returns are at the quarterly frequency.

Unsmoothing produces a dramatic effect. The minimum reported return during the real estate downturn in the early 1990s is –5.3% during the quarter ending December 1991. The corresponding unsmoothed return is –22.6%. During the financial crisis, NCREIF returns reached a low of –8.3% in December 2008. The unsmoothed return during this quarter is –36.3%. The volatility of the raw NCREIF returns is 2.25% per quarter, whereas the volatility of the unsmoothed returns is 6.26% per quarter. This approximates the volatility of stock returns, which is around 7.5% per quarter. Correlation (and hence beta) estimates are also affected by unsmoothing: the correlation of raw NCREIF returns with the S&P 500 is 9.2% and this rises to 15.8% once the unsmoothing correction is applied.

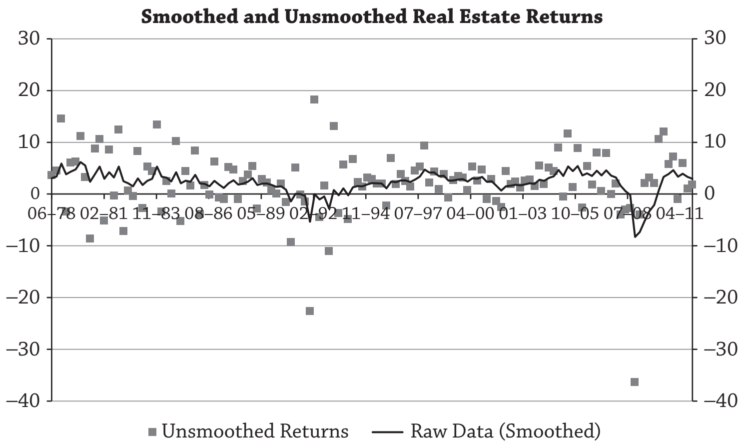

Figure 13.3

[18] Unsmoothing corrections produce similar effects in other illiquid markets. Campbell (2008), for example, estimates that unsmoothing increases the volatility of art market returns from 6.5% to 11.5%.